Prahari: how the digital twin works

Proof of concept for type 2 diabetes · Team Songer, IIM Visakhapatnam · all data synthetic

1 DATA IN

Static EHR (Synthea)

- Age, sex, diagnosis, years since diagnosis
- HbA1c, BMI, eGFR, BP, lipids
- Medicines, comorbidities
- Synthetic genetic risk + family history

Dynamic streams

- CGM glucose every 5 min
- Watch: heart rate, HRV (RMSSD), steps, sleep stages
- Meal log (photo/app), insulin pen

Virtual patient simulator

- 204 people x 21 days, Indian routines, hidden physiology richer than the twin

2 INGEST & PROTECT

Ingestion layer

- FHIR R4 bundles (ABDM-style) for the record
- Time-aligned 5-min grid, gaps flagged
- Pseudonymous IDs, only fields the model needs (DPDP data minimisation)
- Consent + audit trail (design)

Leakage guard

- At time t, use only data before t
- Meals count once logged
- Habits learned in week 1 only
- Patient-level train / validation / test split

3 DIGITAL TWIN

EHR prior

- HbA1c, BMI, diagnosis, medicines set starting values

Onboarding fit (days 1-7)

- 5 personal parameters by MAP: sensitivity, fasting level, meal gain, absorption, dawn

Stay in sync (every 5 min)

- Observer corrects state from CGM, estimates unlogged food

Re-fit every night

- Trailing 7 days, warm start

4 HYBRID AI

Feature store

- CGM history and trend
- Meals, insulin, steps, HR, HRV, sleep
- EHR features
- Twin forecast, peak, parameters

LightGBM models

- Glucose at 30/60/90/120 min
- 80% prediction band (quantiles)
- Spike risk: >180 and >250 in 2 h
- SHAP: top 3 reasons

Evaluation

- Held-out patients, days 8-21
- Ablation, Clarke grid, AUROC, lead time

5 CLINIC

Doctor dashboard

- Patients ranked by spike risk
- 2-hour forecast with band
- Why: top reasons
- Twin parameters over time
- What-if: walk, portion, sleep

Patient nudge

- One heads-up per meal, 120-min cooldown
- Suggested action from the twin (e.g. 15-min walk)
- Doctor sets thresholds

Closed loop. Every new CGM reading updates the twin's state; every night the twin re-fits its parameters; the hybrid model re-scores risk every 15 minutes. Stack: Python 3, NumPy, SciPy, Numba, pandas, LightGBM, SHAP, scikit-learn, Plotly.js dashboard (static, works offline). Not a medical device.